

Adaptive KL Regularization via Meta-Learning

Proposed Solution and Novelty Analysis

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1. Proposed Algorithm: MetaKLController

Background: Standard GRPO Objective

Group Relative Policy Optimization (GRPO) [1] optimizes the following objective:

$$\mathcal{L}(\theta) = \mathbb{E} [A_i \cdot \log \pi_{\theta}(o_i \mid q)] - \beta \cdot \text{KL}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (1)$$

where the group-normalized advantage is:

$$A_i = \frac{r_i - \mu_r}{\sigma_r} \quad (2)$$

and β is a **fixed** scalar hyperparameter set before training begins. The proposed **MetaKL-Controller** replaces this fixed β with a dynamically predicted value β_t at each training step t .

Algorithm

Algorithm 1 MetaKLController: LSTM-Based Adaptive KL Regularization for GRPO

Require: Pre-trained policy π_θ (Qwen2.5-0.5B), frozen reference π_{ref} , training set $\mathcal{D}$ (MATH, 7,119 samples), LSTM controller f_ϕ , initial coefficient $\beta_0 = 10^{-4}$, learning rate $\eta = 5 \times 10^{-7}$

Ensure: Fine-tuned policy π_θ^* with adaptive β scheduling

- 1: **Initialize** $\pi_\theta, \pi_{\text{ref}}$ (frozen), LSTM f_ϕ
- 2: **Initialize** hidden state $h_0, \beta_0 \leftarrow 10^{-4}$
- 3: **for** each training step $t = 1$ to T **do**
// Step (a): Rollout Generation
- 4: Sample batch of 16 prompts $\{q_j\}$ from $\mathcal{D}$
- 5: Generate $n=2$ responses per prompt via vllm at temperature = 1.0
// Step (b): Reward Computation
- 6: Compute reward for each response:

$$r_i = \begin{cases} +1.0 & \text{if response contains correct } \boxed{\cdot} \text{ answer} \\ -1.0 & \text{if response is malformed} \end{cases}$$

- // Step (c): Advantage Estimation (standard GRPO)*
- 7: Compute group-normalized advantage: $A_i = \frac{r_i - \mu_r}{\sigma_r}$
// Step (d): ★ NOVEL STEP — Adaptive β via MetaKLController
- 8: Collect training signal window:

$$\mathbf{s}_t = [\mathcal{L}_t^{\text{KL}}, \bar{r}_t, \sigma_{r,t}, \|\nabla\|_t]$$

- 9: Predict adaptive coefficient via LSTM:

$$\beta_t, h_t = f_\phi(\mathbf{s}_t, h_{t-1})$$

- 10: β_t **dynamically replaces** the fixed scalar β of standard GRPO
// Step (e): Actor Update
- 11: Compute objective:

$$\mathcal{L}(\theta) = \mathbb{E}[A_i \cdot \log \pi_\theta(o_i | q)] - \beta_t \cdot \text{KL}(\pi_\theta \| \pi_{\text{ref}}) + \lambda_{\text{ent}} \cdot \mathcal{H}(\pi_\theta)$$

- 12: Update θ via clipped surrogate gradient (clip ratio = 0.2)
 - 13: Update LSTM hidden state: $h_{t-1} \leftarrow h_t$
 - 14: **end for**
 - 15: Evaluate π_θ^* on MATH500 (greedy decoding, 434 problems)
 - 16: **return** π_θ^* , final MATH500 accuracy
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Complexity: $\mathcal{O}(T \cdot G \cdot L)$ where T = training steps, G = group size, L = LSTM forward pass cost.

Key distinction: Only Step (d) is novel. All other steps are identical to the validated GRPO baseline [2], making the comparison clean and controlled.

2. Novelty of the Proposed Solution

The novelty of MetaKLController is established by showing that no existing work occupies the intersection of: (a) a *learned* (not hand-engineered) adaptation mechanism, (b) targeting the KL coefficient β specifically, and (c) within GRPO for LLM reasoning. The argument is organized in three layers.

2.1 Works That Modify the KL Formulation But Retain a Fixed β

Liu et al. [3] and Shah et al. [4] both address how KL is *computed* — fixing bias in the estimator and gradient formulation — but β remains a static scalar in both works. Yuan et al. [5] go further and replace KL with Bregman divergences entirely, achieving +5.5 points on GSM8K over baseline GRPO. None of these works ask the question this thesis asks: *should β change over time, and can that change be learned?*

2.2 Works That Introduce Adaptive Training Mechanisms But Not Adaptive β

SEED-GRPO [6] adapts update magnitude *per prompt* using semantic entropy, while keeping β fixed globally. GRPO- λ [7] adapts the *reward strategy* per group based on correctness ratios. DisCO [8] enforces KL via constrained optimization but with a fixed constraint boundary. All three are adaptive in spirit but operate on a fundamentally different axis than the MetaKLController.

2.3 Works That Use Learned Controllers for RL Hyperparameters Outside LLM Fine-Tuning

Ali et al. [9] and Occorso et al. [10] both use recurrent meta-learners to tune RL hyperparameters — architecturally closest to the LSTM design proposed here. However, both operate on classical RL tasks and general hyperparameters, not on the KL coefficient specifically within GRPO for LLM reasoning. HyperController [11] uses a Kalman filter (linear assumption) rather than an LSTM, limiting its ability to capture the non-stationary dynamics of LLM training.

2.4 Most Closely Related Work: SAFE

SAFE [12] is the strongest competitor. It dynamically adjusts KL penalties using a PID controller based on entropy state and reward velocity. The critical difference is that SAFE uses **hand-engineered control rules** (PID), while MetaKLController uses a **learned LSTM** that discovers adaptation patterns directly from training signals — without requiring predefined rules. Additionally, SAFE targets PPO, not GRPO.

2.5 Comparative Summary

Table 1: Comparison of MetaKLController Against Related Work

Dimension	MetaKLController	Closest Alternatives
What adapts	β (KL coefficient) over training steps	SAFE: KL threshold via PID [12]; SEED-GRPO: update magnitude per prompt [6]; DisCO: KL via constrained opt. [8]
How it adapts	Learned LSTM observing KL loss, rewards, gradients	SAFE: hand-designed PID rules; SEED-GRPO: semantic entropy heuristic; HyperController: Kalman filter [11]
Domain	GRPO for LLM math reasoning	Ali et al.: general NN hyperparameters [9]; Occorso et al.: TRPO in classic RL [10]
Adaptation granularity	Per training step	SEED-GRPO: per prompt; GRPO- λ : per group [7]

2.6 Novelty Contribution Statement

“MetaKLController is the first learned, recurrent controller that dynamically adapts the KL coefficient β within GRPO specifically for LLM mathematical reasoning, using live training signals — occupying an intersection of adaptive regularization, meta-learned control, and LLM fine-tuning that no prior work addresses.”

Empirical support: The baseline experiment (Section 5 of the progress report) shows that KL loss remained flat near 0.001 across all 444 training steps with fixed $\beta = 10^{-4}$, demonstrating identical regularization strength during early exploration and late convergence phases. This is precisely the inefficiency that MetaKLController is designed to correct, and that no existing method in the reviewed literature addresses within GRPO.

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